18IS54/ Test-1

Question 01 (ES)

a) Construct the transition diagram to recognize the tokens given below i) Relational operation ii) Unsigned number (6) b) Differentiate between an Interpreter and Compiler (4)			*ES*
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	Response template	**Information for graders**	
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Allows a response of a few sentences or paragraphs. This must then be graded manually.			

Question 02 (ES)

a) Define Left recursion and Left factoring and apply the same for the grammar, if required i) E -> E * T/T ii) T -> id + T/id (4) b) Construct the predictive parsing table for the following grammar S - > AaAb | BbBa A -> ε B -> ε (6)	*ES*

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Question 03 (ES)

Given the grammar S -> XS | dS | ε X -> Y | Zb | aY Y -> cZ Z -> e i) Construct FIRST and FOLLOW sets ii) Construct the predictive parsing table iii) Show the moves made by the predictive parser on the input "dace" (10)			*ES*
		Default mark:	10
		Response format:	HTML editor with file picker
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Question 04 (ES)

a) Differentiate between Syntax Analysis and Semantic Analysis with an example. (6) b) Briefly describe the use of compilers in current context. (4)			*ES*
		Default mark:	10
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Question 05 (ES)

a) How can the task of reading the source program be speeded up? Explain the methods. (6) b) Describe the languages denoted by the following regular expressions: with atleast two acceptable input strings for each. i) (a|b)*a(a|b)(a|b) ii) ((ε|a)b*)* (4)	*ES*

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Question 06 (ES)

a) Explain the ways of handling reserved words which look like identifiers while recognizing tokens. (4) b) With an example differentiate between Transition table for NFA and DFA. (6)			*ES*
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		Response format:	HTML editor with file picker
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Question 07 (ES)

a) Divide the following C++ program into appropriate lexemes
float limitedSquare(x) float x {
/* returns x-squared, but never more than 100*/
return (x<=-10.0 || x >= 10.0)?100:x*x;
} (4)
b) Explain with an example simulating NFA. (6)
	ES

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Question 08 (ES)

a) Convert the following regular expressions to deterministic finite automata i) ((ε|a)b*) ii) (a|b)*abb(a|b)* (4) b) With an example show how to minimize the number of states of a DFA. (6)	*ES*

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Question 09 (ES)

a) Show that the following grammar is ambigious E -> E+E | E*E| (E) | id Write an equivalent un-ambigious grammar for the same (6) b) Identify and provide the importance of any four tools used for constructing compilers. (4)	*ES*
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Question 10 (ES)

Explain McNauthton-Yamada-Thompson algorithm with an example. (10)	*ES*

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Question 11 (ES)

Implement a scanner that adds line numbers to text, sending its output to the screen/or a file. (10)			*ES*
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Question 12 (ES)

a) Consider the grammar exp -> exp op exp | (exp) | *number* op -> + | - | * Is the grammar ambiguous? If so, show that the grammar is ambiguous with the help of parse tree for input string. (6) b) What are the considerations for error recovery in top down parsers. (4)	*ES*

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Question 13 (ES)

a) Given the grammar A -> (A) A | ε i) Construct First and Follow sets for the terminal A ii) Show this grammar is LL(1) (6) b) Consider the following grammar of simplified C declarations declaration -> type var-list type -> *int* | *float* var-list -> *identifier*, var-list | *identifier* i) Left factor the grammar ii) Construct First and Follow sets for the non-terminals (4)			*ES*
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Question 14 (ES)

a) Explain the process of adopting panic mode recovery in LL(1) Parsers. (6) b) Explain the process of constructing LL(1) parsing table. (4)			*ES*
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Question 15 (ES)

a) Consider the following grammar stmt-sequence -> stmt stmt-seq' stmt-seq' -> ; stmt-sequence | ε stmt -> *s* i) Construct the First and Follow sets for the grammar ii) Construct the LL(1) parsing table (6) b) Given the grammar exp -> exp addop term | term addop -> + | - term -> term mulop factor | factor mulop -> * factor -> *(*exp*)* | *number* Write down leftmost derivations, parse trees for the following expressions: i) 3+4*5-6 ii) 3*(4-5+6) (4)	*ES*

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Question 16 (ES)

Write a grammar for Boolean expressions that includes the constants *true* and *false*, the operators *and*, *or* and *not*, and parentheses. Be sure to give *or* a lower precedence than *and* and *and* a lower precedence than *not* and to allow repeated *not*'s, as in the Boolean expression *not not true*. Also be sure your grammar is not ambiguous. (10)	*ES*
Default mark:	10
Response format:	HTML editor with file picker
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Question 17 (ES)

a) Consider the following grammar representing simplified LISP-like expressions: lexp -> atom | list atom -> number | identifier list -> (lexp-seq) lexp-seq -> lexp-seq lexp | lexp Write a leftmost and a rightmost derivation for the string *(a 23 (m x y))* (4) b) stmt -> if-stmt | *other* if-stmt -> *if (*exp*)* stmt | *if (*exp*)* stmt *else* stmt exp -> *0* | *1* Provide proof to prove that the above grammar is ambiguous. (6)	*ES*
Default mark:	10
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Question 18 (ES)

a) Explain the process of converting NFA to DFA with an example. (6) b) Explain the process of optimizing DFA with example. (4)	*ES*

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Question 19 (ES)

a) Explain the process of conflict resolution in Lex through an example. (4) b) Explain the architecture of a transition-diagram-based lexical analyzer. (6)			*ES*
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Question 20 (ES)

A grammar is ε-free if no production body is ε i) Give an algorithm to convert any grammar into an ε-free grammar that generates the same language ii) Apply your algorithm to the grammar S -> aSbS | bSaS | ε. (10)	*ES*

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